

Adaptive local neighborhood search and dual attention convolution network for complex semantic segmentation towards indoor point clouds

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ABSTRACT

In indoor scenes, due to the various semantic categories and scale differences of objects, the complex spatial structure of objects, and the different reflection characteristics of laser resulting in the uneven density distribution and the complex and diverse detail features of the point clouds, making the semantic segmentation of point clouds still a challenging task. In this paper, we propose an adaptive local neighborhood search and dual attention convolution network (ADNet) for complex semantic segmentation towards indoor point clouds. First, an adaptive local neighborhood search (ALNS) mechanism is constructed to address the problem that the traditional KNN is difficult to adapt to the changing density of complex point clouds. This mechanism dynamically adjusts the neighborhood radius according to the point cloud density, allowing for the construction of graphs that enable the network to learn local features with optimal receptive fields. Additionally, we propose a multi-scale feature fusion module to effectively integrate features of different scales by utilizing dynamic context-aware attention convolution (DCAA-Conv) to capture key context information between distant points and graph space attention convolution (GSA-Conv) to capture complex spatial geometric relations between nearby points and avoid the interference of noisy point clouds. Finally, spatial pyramid pooling specialized for indoor point clouds (SPP-IPCs) is employed for multi-scale feature fusion to enhance the ability to capture and express features at various scales. The proposed ADNet demonstrates proficiency in identifying minor categories and distinguishing building structures within complex environments, achieving superior segmentation accuracy compared with existing work on ScanNet v2 and S3DIS datasets, thus providing a new technical approach for accurate and complex 3D scene analysis.

1. Introduction

The development of 3D scanning technology has led to more efficient and convenient acquisition of indoor point cloud data (Yue et al., 2024). This provides the possibility of intelligent processing and automatic understanding of indoor scenes. In this context, indoor point cloud semantic segmentation becomes one of the key technologies for scene understanding. It aims to recognize and extract key objects from complex indoor environments by transforming these disordered high-density point clouds into structured semantic information, thus providing a basis for applications such as indoor navigation (Liu, Wei et al., 2022; Thomas et al., 2021), scene reconstruction (Hu et al., 2023), and virtual reality (Wirth et al., 2019). The semantic information refers to the category labels of the actual physical objects or

structures to which each point in the point cloud belongs, such as walls, floors, columns, etc.

Traditional geometric feature-based methods (Ji et al., 2019; Xiang et al., 2018; Zhang et al., 2016; Zhao, Yue et al., 2023), although fast in the training of semantic annotation, are often difficult to adapt to complex and changing indoor environments due to the incompleteness of the features. Therefore, researchers have begun to explore methods that incorporate deep learning techniques to provide new solutions for the semantic segmentation of point clouds. Among them, PointNet (Qi, Su et al., 2017) and its improved methods (Jiang et al., 2023; Lei et al., 2022; Qi, Yi et al., 2017; Zhao et al., 2019) are especially prominent. Although these point-based methods can directly process and learn the global features of each point independently, they still have two

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problems when faced with the overall understanding of indoor environments. One is that these methods are prone to information loss when dealing with small objects and subtle complex structures, and the other is that these methods cannot adequately capture long-range dependencies in point clouds.

To overcome the above issues, graph convolution-based methods have emerged. Graph convolution networks can effectively capture local and global structural information of indoor point clouds (IPCs) by directly performing convolution operations on the graphs. At the same time, such methods can fully take into account the complex spatial relationships among points and convey the information on a global scale. For example, Landrieu and Simonovsky (2018) propose the SPG model to realize semantic segmentation by constructing the superpoint graphs of the point clouds through the directed attribute graph. Regularized graph CNN (RGCNN (Te et al., 2018)) adaptively captures the graphs of the point clouds by updating the Laplace matrix at each layer of the network. A double filter graph convolution network (DFGCN (Li et al., 2020)) uses the coordinate information and normal vector of each point to construct a fully connected graph. It is easy to realize that constructing a high-quality graph may directly affect the performance of the network.

Although many point cloud semantic segmentation methods have been investigated for indoor scenes, there are still some shortcomings in capturing local details as well as multi-scale feature fusion. Specifically, existing methods cause detail loss when capturing small but important indoor structures or objects. In addition, the relationship between different scale features in the point cloud is not efficiently considered. Therefore, how to determine the connection relationship among points and how to define the weights of detail features is a critical issue for the semantic segmentation of IPCs.

To address the above problems, we propose a novel semantic segmentation algorithm for IPCs. This approach not only enhances the accuracy of recognizing objects and details of different sizes; it improves the ability of the network to understand indoor scenes. We first introduce an adaptive local neighborhood search (ALNS) mechanism, which can dynamically construct the graph based on the local features of the IPCs, thus capturing the local structure information more accurately. Then, we propose a multi-scale feature fusion module based on the attention mechanism. The module combines dynamic context-aware attention convolution (DCAA-Conv), graph space attention convolution (GSA-Conv), and spatial pyramid pooling specialized for IPCs (SPP-IPCs), which can effectively extract and fuse features at multiple levels in IPCs. To be specific, DCAA-Conv can dynamically adjust its weights based on the contextual information of each point in the indoor space to better capture local and global features. GSA-Conv is more effective in capturing complex spatial relationships than traditional convolutional operations, leading to a better understanding of the connections and interactions between points. SPP-IPCs enables the generation of feature representations with different receptive fields, enabling the network to focus on both large-scale global information and small-scale local details.

Our approach places special emphasis on effective graph construction for IPCs and the capture of multi-scale features such as local details, global context, and spatial structure in indoor environments. These key steps are essential for improving the accuracy of semantic segmentation of IPCs. We conducted tests on representative ScanNet v2 and S3DIS datasets. By comparing with other algorithms, the advantages of the method in indoor scene parsing are verified. In short, the main contributions of our approach are summarized as follows:

- This paper focuses on the semantic segmentation of point clouds for indoor scenes, aiming to solve problems such as loss of details and interference from additional irrelevant point features caused when extracting features with uneven density distribution. The segmentation accuracy for minor categories and confusing indoor building structures is improved by extracting rich multi-scale features.
- We design an ALNS mechanism that uses the similar feature relationship among distributed non-uniformly points to dynamically adjust the radius of the neighborhood, effectively addressing the limitation of fixed neighborhood radius that makes it difficult to obtain complex local details.
- We design a multi-scale feature fusion module that aggregates local detail features, contextual semantic features, and spatial geometric features. This module comprises three main components, i.e.,
 - DCAA-Conv is proposed to enable the network to adapt to intricate geometric structures in indoor scenes and prevent the loss of crucial context features.
 - GSA-Conv allows the network to flexibly capture complex spatial relationships within IPCs.
 - SPP-IPCs facilitates hierarchical integration of contextual semantic information and spatial structure details from different receptive fields, thus enhancing the representation of the features for various regions in indoor scenes.

The remainder of this paper is organized as follows. Related works are reviewed in Section 2. Section 3 introduces the novel methods for indoor point cloud semantic segmentation in detail. Section 4 presents our experimental results and compares them with other methods. Finally, our conclusions and future work are provided in Section 5.

2. Related works

In this section, we mainly provide an overview of two methods to point cloud semantic segmentation: the graph convolution-based method and the attention mechanism-based method, and present some recent research work on the task of semantic segmentation in indoor scenes.

2.1. Graph convolution-based methods

As graph convolutional neural networks (GCNs) have demonstrated excellence in the field of non-Euclidean data processing, many works have incorporated the technique into point cloud analysis. By constructing the graph of point clouds, the relationships among points can be captured more efficiently and the semantic feature extraction of the point clouds can be improved. Wang et al. (2019) first propose to utilize EdgeConv instead of MLP to learn the global shape features while acquiring the local neighborhood information. The dynamic graphs constructed by DGCNN can achieve updates at each layer of the network, thus continuously constructing new graphs. For improving the efficiency of graph convolution applied in point cloud semantic segmentation and reducing the parameters of the model, multi-scale dynamic GCN (MSDGCN (Zhai et al., 2020)) first obtains a sparse point cloud dataset using the farthest point sampling and k-nearest neighbor (KNN); and then extracts and aggregates local features among the center point and its neighbors using EdgeConv. Khan et al. (2020) propose a feature-based graph convolutional network (FGCN) that encodes 2D features with 3D spatial arrangement as an undirected symmetric graph and generates a complete segmentation graph by multi-layer local graph convolution. The hierarchical depth-wise graph convolutional neural network (HDGCN (Liang et al., 2019)) is composed of a hierarchical structure consisting of a series of DGConv blocks, which can extract both local and global features of the point clouds. In addition, the design of depth-wise graph convolution can effectively alleviate the memory consumption of the network compared to FGCN. The graph convolution-based dual fusion network (GDFNet (Lu et al., 2024)) enhances feature mining of point clouds with uneven density and improves the granularity of critical local information by utilizing density features and spatial structure information.

Recently, multi-scale features and multi-view learning have been applied to state-of-the-art networks as a way to enhance the abilities

Table 1
Summary of methods for indoor scenes.

Methods	Brief description
PGCNet (Sun et al., 2020)	Surface patches are used as efficient representations of IPCs and semantically labeled with delineated indoor layouts and objects.
GCN-MLP (Wang et al., 2021)	The dense connectivity between different feature maps is used to extract point cloud features in indoor scenes.
NAS (Lin et al., 2021)	Point-by-point features are enhanced by communicating between multiple levels of features to provide more information on semantics and localization.
LGGCM (Du et al., 2022)	The long- and short-range dependence on the point cloud is achieved, and a smooth module is introduced to improve the robustness of the network to noise.
Liu, Lu et al. (2022)	Improving segmentation efficiency by automatically learning the importance of feature channels.
GTNet (Zhou, Wang et al., 2024)	Differences and correlations among points are considered and both local and global features of the point cloud are learned.
ADNet (Ours)	Efficient segmentation of IPCs with different densities and multi-scale feature fusion of local details, global context, and spatial geometry.

of models to understand the topology of point clouds and perceive complex scenes. A hybrid semantic affinity learning method (HSA (Song et al., 2021)) is designed by combining semantic information with high-level prior knowledge. From a global perspective, the method captures the internal structural associations of the point clouds through a graph convolution layer and makes category determination following prior knowledge. The method solves the semantic obfuscation problem through a hybrid affinity learning strategy and can be extended to instance segmentation in the future. To reduce the differences in semantic features between different levels, the feature graph convolution network with attentive fusion (FGC-AFNet (Chen et al., 2023)) continuously expands the receptive fields to acquire and fuse features at multiple scales. An adaptive multiview graph convolutional network (AM-GCN (Niu et al., 2024)) combines local features under multiview projection and global features of point clouds to provide a technical basis for the field of indoor robot grasping. This method changes the previous traditional graph construction method through multi-view graph construction, which also provides new ideas for our research.

Most of the methods mentioned above use KNN to construct graphs. This way is difficult to deal with dense and sparse regions. It is easy to ignore the detailed information of the local structure, which increases the difficulty of extracting global context features. However, our ALNS mechanism is specially designed to deal with the problem of uneven density of IPCs. Dynamically adjusting the radius of the neighborhood ensures that enough local details can be captured in the sparse area and that key features can be effectively extracted in the dense area to improve efficiency.

2.2. Attention mechanism-based methods

In graph convolution operations, the incorporation of the attention mechanism dynamically adjusts the feature aggregation weights based on the relationship between the features of each point and its neighbors. This allows the network to focus more on local regions with important semantic information and better utilize the detailed information of local structures. The simultaneous focus on contextual information enhances the expressiveness of the features, thus enabling the model to handle large-scale point clouds more efficiently.

Wang et al. (2019a) propose a graph attention convolution (GAC) that selectively focuses on the most relevant parts by assigning attention weights to different neighbors to achieve fine-grained segmentation and avoid feature mixing between objects. Hu, Zhang et al. (2020) introduce an attention mechanism to capture semantic correlation from the point clouds and use multi-scale supervision to guide contextual aggregation. In contrast to the mentioned traditional attention-based approaches, an adaptive graph convolution (AdaptConv (Zhou et al., 2021)) is proposed to generate adaptive convolution kernels based on dynamically learned features to effectively capture the diverse relationships between points across different semantics. Similarly, the augmented edge representation module in a dual attention neural

network (DA-Net (Zhang et al., 2022)) can augment hidden geometric features by adaptively representing local positions and orientations. The network also utilizes the elevation attentive pooling module to further encode discriminative local features. The method provides better access to local topology and long-range dependencies among points, which improves object-level and scene-level segmentation accuracy. A multilevel attention-based U-shape graph neural network (MAUGNN (Zou et al., 2020)) effectively learns and fuses multilevel features and adaptively fuses coarse- and fine-grained features through an attention-based connection mechanism. A dynamic graph attention network (DGAT-net (Miao et al., 2024)) captures local details and global dependencies of point clouds by introducing a spatial self-attention mechanism and a graph attention mechanism. The network improves the understanding of complex spatial relationships while reducing computational complexity, and achieves superior performance in semantic segmentation.

Although the above networks can consider the structural detail information of the point cloud more comprehensively than the standard graph convolution methods, they still have some limitations in complex scenarios, especially in the multi-scale feature extraction and effective fusion of the point clouds. Unlike these networks, ADNet exploits a multi-scale feature fusion module that can simultaneously extract context-aware information and spatial structural relationships. The extracted finer-grained features are particularly important for the perception and recognition of complex structures and object boundaries in indoor environments.

2.3. Methods for indoor scenes

Point cloud semantic segmentation can label and recognize different objects or regions in an indoor scene, thus providing accurate scene understanding for applications such as indoor navigation and virtual reality. Currently, many approaches have begun to utilize the concepts of deep learning and GCNs to address the complex semantic segmentation toward IPCs. Sun et al. (2020) propose a patch graph convolution network (PGCNet) to efficiently represent large-scale IPCs by utilizing surface patches. In addition, this network uses dynamic edge convolution operations to continuously update the graphs to encode hierarchical edge features. The method can be effectively used for indoor space layout and object recognition. Wang et al. (2021) design a method for indoor point cloud segmentation by combining GCN and MLP. A point cloud channel aggregation module is introduced into the method to enhance the representation of distinguishing features in IPCs by aggregating depth features at multiple levels. To ensure that the multi-level features are adequately fused, Lin et al. (2021) propose a method-neural architecture search (NAS) - specifically designed to search for feature pyramids applicable to IPCs. We are inspired by this method, so a novel SPP-IPCs can be used in ADNet to achieve multi-scale feature fusion as a way to improve semantic segmentation accuracy for IPCs.

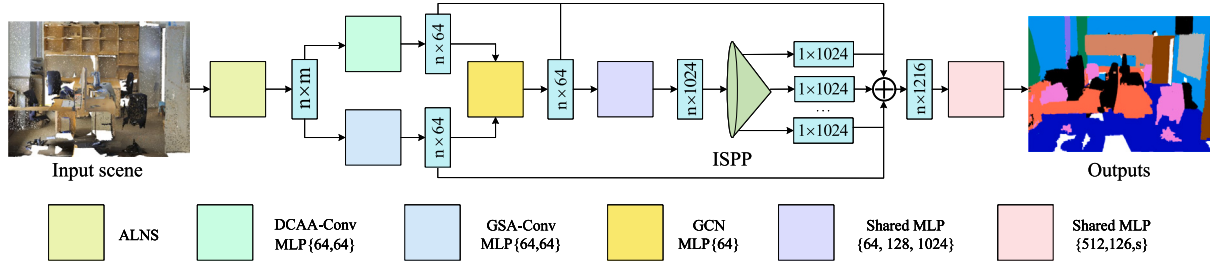


Fig. 1. Architecture of the proposed ADNet.

Recently, many new methods have successfully applied attention graph convolution to the task of semantic segmentation of IPCs. Du et al. (2022) propose a local-global graph convolutional method (LGGCM), which can efficiently capture the long- and short-range dependencies and global contextual information of the point clouds. The core of the method lies in applying local spatial attention convolution to extract the spatial geometric features of the point clouds and enhancing the robustness to noise through a smoothing module. Liu, Lu et al. (2022) enable the network to automatically learn the importance of the feature channels of IPCs by introducing a channel attention module. The network improves the segmentation accuracy only in some of the categories, but the boundary segmentation between walls and other objects still needs to be improved. Graph Transformer (GTNet (Zhou, Wang et al., 2024)) is proposed to learn the features of local neighbors instead of limiting to global attention feature extraction. Therefore, in the indoor semantic segmentation tasks, this method can effectively utilize the variability and correlation between points to improve the prediction ability.

These above methods effectively preserve and extract the structural and semantic information of IPCs by introducing techniques such as attention mechanism, graph convolution, and feature pyramid. However, there is still room for development and improvement in handling complex point cloud data with different densities and balancing local and global features. Unlike the above networks, ADNet can handle complex point clouds with different densities and utilize SPP-IPCs for multi-scale feature fusion. The objective is to enhance the learning of local details, global context, and spatial geometric features of IPCs to improve the global understanding and boundary sensing of indoor structures. To improve readability, this paper summarizes the methods for IPCs and the proposed ADNet in Table 1.

3. ADNet

We propose a network suitable for complex semantic segmentation toward IPCs with the structure shown in Fig. 1. First, a graph convolutional neural network is used to process large-scale indoor point clouds, in which the ALNS mechanism is employed to construct the graphs of point clouds. Then DCAA-Conv and GSA-Conv are used to obtain the key features of the point clouds, respectively, and the multi-scale features are aggregated by SPP-IPCs. The proposed neighborhood search mechanism and multi-scale feature fusion module enable the whole network to improve the segmentation accuracy to a certain extent. The details are elaborated in Sections 3.1 and 3.2.

In this network, the input is an arbitrary point cloud $P = \{p_i \in \mathbb{R}^m\}_{i=1}^n$ of n points with the neighborhood of a point p_i as $\mathcal{N}(p_i)$. These points may contain m -dimensional attributes such as coordinates (x_i, y_i, z_i) , colors (r_i, g_i, b_i) , or normal vectors (v_{xi}, v_{yi}, v_{zi}) , and are denoted as the point cloud containing only 3D coordinate information when $m = 3$. Furthermore, m can also be denoted as the feature dimension of a given network layer in GCNs.

3.1. Adaptive Local Neighborhood Search (ALNS)

Most of the existing point cloud semantic segmentation algorithms rely on KNN and ball query (Huang et al., 2024; Yin et al., 2021)

to search the neighborhood structure of the point clouds. In addition, some algorithms also employ Octree (Chen et al., 2021; Lei et al., 2019) to spatially index large-scale point clouds to reduce retrieval time and memory consumption. However, these neighborhood search strategies are difficult to determine appropriate neighborhood scales based on different density regions when applied to semantic segmentation of IPCs, which tends to cause loss of details in sparse areas and increase the computational cost in dense areas; and ignores certain spatial structural information, which leads to boundary effects. Therefore, we first analyze the relative advantages and disadvantages of these conventional search strategies, and then propose a novel ALNS strategy for improving their shortcomings as follows.

Conventional search strategies. KNN is the simplest and most intuitive neighborhood exploration method, which requires traversing each point and calculating the distance. Therefore, the computational complexity is high and time-consuming when dealing with large-scale point cloud data. Moreover, in the segmentation of indoor scenes, the choice of k is more sensitive to other parameters and lacks adaptivity. Ball query is to find other points p_j in its neighborhood $\mathcal{N}(p_i)$ within a specified radius r centered on each point p_i according to the following formula:

$$p_j \in \mathcal{N}(p_i), \quad \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2} < r. \quad (1)$$

Since this strategy takes into account the difference in distance and density, it can handle non-uniformly distributed points. However, it also suffers from the same problems of inefficiency and unstable segmentation results when dealing with large-scale IPCs and the selection of the radius range. To efficiently process large-scale point clouds, Octree provides a hierarchical representation of the space by constructing a hierarchical tree structure that provides access to local regions. However, Octree is more complicated to construct and maintain, especially for non-uniformly distributed points, which may lead to uneven spatial division.

ALNS. ALNS is a method for graph construction that dynamically adapts to different density regions in the semantic segmentation of IPCs. It can dynamically adjust the neighborhood scale of each point. The neighbors of each central point are determined by capturing the local density and location information of the points. In this way, it solves the problems of local detail loss and insufficient exploitation of spatial features of traditional methods when dealing with IPCs with uneven density. Local spatial graphs $G(V, E)$ are constructed based on spatial relationships among the central point and its neighboring points, where $V = \{1, 2, \dots, n\}$ denotes a finite set of n vertices and $E \in |V| \times |V|$ denotes the set of edges, that is, the edges from all neighboring points p_j to the center point p_i in the neighborhood $\mathcal{N}(p_i)$. They are dynamically updated based on feature weights in regions of varying density, as shown in Fig. 2.

As shown in Fig. 3, for dense regions of the point cloud, a smaller neighborhood scale can be used to capture local subtle features more accurately, while for sparse regions of the point cloud, a larger neighborhood scale can be used to ensure a wider coverage of information. In this way, we can achieve better local feature extraction and analysis results in IPCs of varying density and complexity.

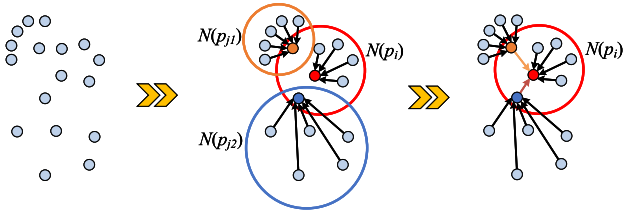


Fig. 2. Local spatial graph construction. The neighborhood radius is adaptively adjusted according to the sparsity of the point cloud, where colors represent different weights.

Specifically, we take the average density of the whole point cloud as a global density threshold δ_{th} and set an initial neighborhood radius R_i : if the number of points in the neighborhood is greater than or equal to δ_{th} , it is a dense region; if the number of points in the neighborhood is less than δ_{th} , it is a sparse region. In ALNS, the neighborhood radius is first dynamically adjusted in the point clouds with different densities, and the number of k nearest neighbors in the neighborhood is selected to represent the local density $D(p_i)$ of each point, where k is not a fixed value:

$$D(p_i) = |KN(p_i)|. \quad (2)$$

The weight $W(p_i)$ of each point in the feature extraction is then calculated based on the $D(p_i)$ of the point cloud, where $\bar{P} = \{p_j \in \mathbb{R}^m\}_{j=1}^k$ denotes the k neighbors of p_i :

$$W(p_i) = \frac{D(p_i)}{\sum_{j=1}^k D(p_j)}. \quad (3)$$

Finally, the weight of each point and the distance $d(p_{ij})$ between the points are used to determine the final neighborhood scale $\mathcal{N}(p_i)$:

$$\mathcal{N}(p_i) = \sum_{j=1}^k W(p_i) \cdot d(p_{ij}). \quad (4)$$

Overall, whether it is the value of k in KNN, the radius in the ball query, or the segmentation threshold in Octree, the selection of these parameters requires different attempts and experiences to achieve the best segmentation results. In dealing with the boundary points at the same time, these three traditional methods may face the problem of discontinuity, resulting in neighborhood search with boundary effect. In contrast, ALNS has the following advantages:

- Dynamic adjustment of local neighborhood radius according to changes in IPCs density. ALNS enables the neighborhood to better adapt to different data distributions and density variations based on the features and contextual information of each point.
- Avoiding loss of local details in areas of uneven density. Through the adaptive mechanism, the parameter values can be adjusted to adapt to regions with uneven density, thus avoiding over-reliance on neighboring information while preserving local features.
- Comprehensive understanding of the 3D structure in indoor spaces. ALNS can understand the spatial structure of the data in a more comprehensive way to extract richer and more accurate feature information.

Based on these, we conclude that ALNS can be applied to construct the graphs of IPCs. However, even with the adaptive approach to dynamically adjust the number and location of neighborhood samples, there are still two problems. One is that certain regions may be overemphasized or ignored, which leads to inconsistent processing of surrounding environment information in different locations. The other is that the graphs constructed by ALNS are relatively complex and some of the neighborhoods are processed too finely, which may reduce the generalization ability of ADNet. To overcome the above problems,

we introduce a multi-scale feature fusion module and use a common regularization technique as described in Sections 3.2 and 3.3.

3.2. Multi-scale feature fusion module

As shown in Fig. 4, the module consists of three main neural units: DCAA-Conv, GSA-Conv, and SPP-IPCs. With specific details, DCAA-Conv first captures semantic relations and relevance of features by learning contextual information of locations at different scales. Then, global spatial relations are extracted as auxiliary inputs using GSA-Conv. Finally, all the key feature maps acquired are fed into the SPP-IPCs module to form a global feature vector for multi-scale feature fusion and representation.

Dynamic Context-Aware Attention Convolution (DCAA-Conv). Traditional convolutional operations (Thomas et al., 2019; Wu et al., 2019) can only locally perceive and extract input features at a fixed scale. In contrast, DCAA-Conv, by introducing an attention mechanism, can dynamically adjust the weights of key features according to the correlation between features, thus enabling the model to understand more fine-grained features and semantic associations.

The local detail features captured by the ALNS module are given as $F_i\{f_i^1, f_i^2, \dots, f_i^j\}$. The convergence of these features indicates that the neighbors p_{jk} that can have an impact on the central point p_i only within an appropriate range have been successfully selected, thus assuring that key features can be extracted next from the rich and complex environment.

First, for each p_{jk} and p_i , the feature similarity score $e(k, n)$ is calculated according to the distance definition function (Euclidean distance):

$$e(k, n) = f_e(\|d'(p_i) - d'(p_{jk})\|), \quad (5)$$

$k = 1, \dots, N, n = [p : p \in j_1, \dots, j_N],$

where k represents the number of nearest neighbors to be considered, n is used to determine the number of nearest neighbors to be considered in the appropriate range, and d' represents the coordinate information of the points.

Next, we use the *softmax* function to normalize $e(k, n)$ into an attention score α_j^n :

$$\alpha_j^n = \frac{\exp(e(j, n))}{\sum_{i=1}^N [p=i]}, \quad (6)$$

where $[p=i]$ represents an indicator function that is used to filter the association between the corresponding neighbors p_{jk} and the current center point p_i . When one of p_{jk} satisfies $[p=i]$ (that is associated with p_i), then $[p=i] = 1$, otherwise $[p=i] = 0$. This weight is used to dynamically adjust the contextual features of the different layers, reflecting the importance of the different areas in the overall semantic understanding.

Finally, α_j^n is applied to the features corresponding to each neighbor to perform weighted fusion and generate context-aware features $\tilde{F}_i$:

$$\tilde{F}_i = \sum_{j=1}^n \left(\alpha_j^n F_i + \left(1 - \frac{k}{m}\right) M_h(s(i)) \right), \quad (7)$$

where m denotes the total number of points that can be sampled as neighbors. $s(i)$ is a function that specifies the location of the central point. $M_h()$ denotes the auxiliary information introduced (e.g., the spatial relationships mentioned in Section GSA-Conv) to further enhance the determination of the model concerning the key targets. h is a collective name for the parameter attributes, such as the 3D coordinates. $\tilde{F}_i$ will be aggregated with other features using SPP-IPCs, such as local details. ADNet then fully utilizes these enhanced and enriched feature representations to achieve higher precision semantic segmentation of indoor point clouds.

The attention weights of DCAA-Conv can be dynamically assigned based on the features in its neighborhood to effectively utilize contextual information and improve computational efficiency. Especially



Fig. 3. Dynamic adjustment of local neighborhoods. The dense regions need to decrease the initial neighborhood radius R_i to maintain high accuracy; the sparse regions need to increase R_i to ensure that enough points are covered.

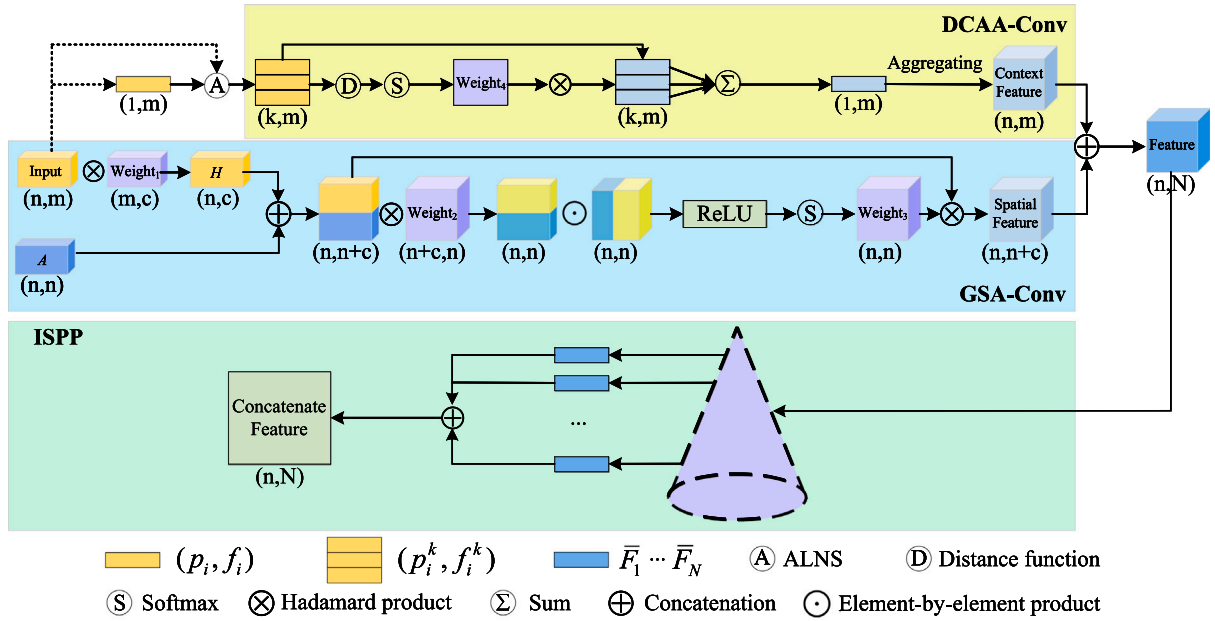


Fig. 4. Multi-scale feature fusion module.

when dealing with indoor scenes with complex semantic relationships, DCAA-Conv can provide additional feature relevance to enable the network to adaptively learn the most valuable features in each point.

Graph Spatial Attention Convolution (GSA-Conv). GSA-Conv can pay more attention to the neighbors p_j that are closer to the current point p_i in terms of spatial distance through the computation of attention weights, thereby capturing the spatial relationship between the points in the region. Therefore, it facilitates the whole network to extract rich feature representations. The design steps of this module are as follows.

Set the input parameters. Given the point clouds containing n points and each point has its own m -dimensional features (position, color, etc.), let the features of these points form an $n \times m$ dimensional feature matrix $\mathbf{X}$, and the points form an $n \times n$ dimensional adjacency matrix $\mathbf{A}$. We take both $\mathbf{X}$ and $\mathbf{A}$ as inputs to the GSA-Conv module.

The feature matrix $\mathbf{X}$ is then mapped to $\mathbf{H} = \mathbf{X}\mathbf{W}$ by a linear transformation, where $\mathbf{W}$ is a weight matrix with dimension $m \times c$ and c denotes the transformed feature dimension.

The weights assigned between each neighbor p_j and the center point p_i indicate the degree of similarity of their features. Methods such as distance metrics are usually used to take relative spatial locations into account so that neighbors receive higher attention weights. For example, a higher physical proximity between a p_j in the $N(p_i)$ and the p_i in the IPCs indicates a stronger correlation. To deal with neighborhoods

of different sizes at different points, we use the *LeakyReLU* function to obtain the similarity score matrix and use the *softmax* function to normalize the annotation to obtain the attention weight matrix α :

$$\alpha = \sigma(\text{LeakyReLU}((\mathbf{H}\mathbf{W}^T) \odot \mathbf{H}\mathbf{W}^{T'})), \quad (8)$$

where $\odot$ denotes element-by-element multiplication and $\mathbf{H}\mathbf{W}^{T'}$ denotes the transpose matrix of the $\mathbf{H}\mathbf{W}^T$. Compared with activation functions such as *ReLU* and *Sigmoid*, *LeakyReLU* has better results in filtering complex similarity relations and reducing redundant information and can alleviate the problem of gradient vanishing.

Eventually, we obtained a global feature matrix $\mathbf{Z} = \alpha\mathbf{H}$ related to spatial location information by weighted aggregation of more attentively focused features.

Compared with traditional local spatial coding methods, GSA-Conv is able to flexibly handle irregular IPCs, enhancing important spatial structures and detailed features in different regions, thus allowing ADNet to adapt to objects with different shapes and sizes as well as more complex indoor environments.

Spatial Pyramid Pooling Specialized for IPCs (SPP-IPCs). Pooling is a common feature aggregation operation, usually dominated by maximum pooling and average pooling. For example, in the DGCNN, the features in $n \times 1024$ dimensions are first pooled using maximum pooling to obtain a global feature in 1×1024 dimensions. Then this feature is copied with the categorical vector to n points in the point

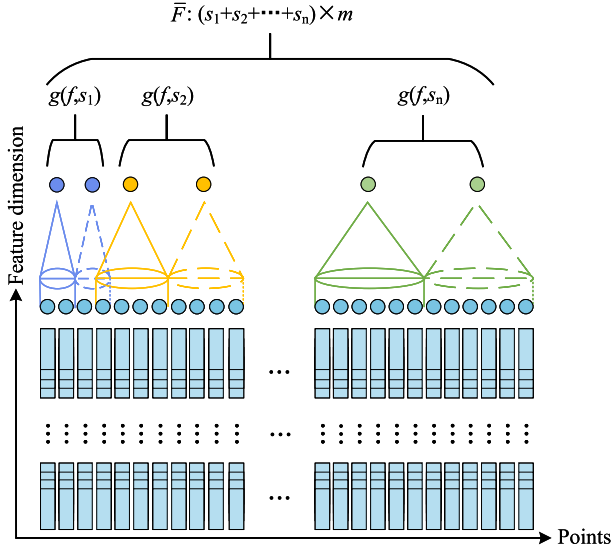


Fig. 5. Indoor spatial pyramid pooling (SPP-IPCs). Given an input of $n \times m$ point cloud features, n represents the number of point clouds, and the blue-striped blocks indicate the m -dimensional features for each point. The cones of different colors represent pooling windows of size s_i . Overlapping pooling occurs when $s_i > stride$. The final output is a multiscale feature of dimension $m \times \sum_{i=1}^n s_i$.

clouds. Finally, the segmentation output scores of n points are obtained after multiple fully connected layers.

However, in the task of complex semantic segmentation of point clouds in indoor scenes, ordinary pooling operations may result in loss of details and degradation of spatial resolution. Therefore, inspired by previous research work (Wang, Wang et al., 2023; Zhang et al., 2021), this module chooses SPP-IPCs to aggregate the features of point clouds at different spatial scales.

The essence of SPP is multi-layer maximum pooling, by adapting feature maps of different sizes, and ultimately only outputs feature information of fixed dimensions. In contrast, the multi-window and overlapping pooling in SPP-IPCs can obtain features related to the dimensions of the input IPCs and enhance the continuity of the features, as shown in Fig. 5. SPP-IPCs can also be formulated as:

$$\bar{F}(p_1, \dots, p_N) = \text{MLP} \left(c(g(f, s_1), \dots, g(f, s_n)), s_i > stride, \right) \quad (9)$$

where f represents the feature map containing $F_i \{f_i^1, f_i^2, \dots, f_i^j\}$, $\tilde{F}_i$, and spatial geometry features. s_i denotes the window size of pooling and is larger than the pooling step, g denotes the pooling operation, and c denotes the fusion of the multi-scale features obtained from SPP-IPCs.

To summarize, our multi-scale feature fusion module aggregates the local detail information, context-aware features, and spatial structural details of IPCs by SPP-IPCs, which enhances the feature representation ability of the network in complex scenes. The module is introduced into indoor point cloud semantic segmentation networks to compensate for the problems caused by the fixed scaling range of a single receptive field. It also can preserve effectively important detailed features in different regions.

3.3. Dropout

Dynamically adjusting the neighborhood scale may increase the risk of overfitting the model to some extent. To reduce the complexity of the model and improve its generalization ability, dropout is used in the network to constrain the parameter space. Specifically, dropout is applied randomly in each training iteration to increase sample diversity and further reduce overfitting. Specifically, as shown in Fig. 6, dropout (Srivastava et al., 2014) is applied to discard half

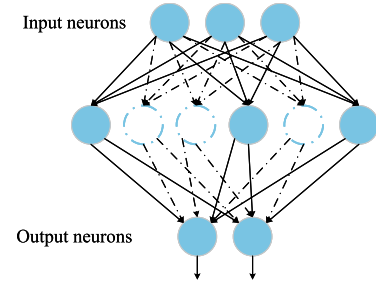


Fig. 6. Dropout. In the process of graph convolutional neural network training, for a particular layer of the neural network in an iteration, half of the neurons (dashed circles) are randomly selected and temporarily discarded, and then the current training and optimization are carried out, and this process is repeated iteratively until the end of training.

of the hidden neurons randomly in each training iteration to increase sample diversity. In indoor point cloud semantic segmentation, dropout prevents ADNet from overfitting the training data by reducing the co-adaptation between neurons. In addition, cross-validation techniques need to be rationally applied to select the optimal dropout probability to avoid the bias problem associated with relying only on training and validation set partitioning.

4. Experiments

In this section, the proposed network is evaluated on challenging and representative indoor point cloud datasets: ScanNet v2 and S3DIS. ADNet can identify and segment the semantic categories of each point in the new indoor scenes after extracting and learning the multi-scale features of point cloud data labeled with semantics in the training. We mainly employ three evaluation metrics to quantitatively analyze and compare the performance of different methods in the semantic segmentation of IPCs: mean Intersection over Union (mIoU), overall accuracy (OA), and mean Accuracy (mAcc). They are computed as:

$$mIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{p_{ii}}{\sum_{j=0}^k p_{ij} + \sum_{i=0}^k p_{ji} - p_{ii}}, \quad (10)$$

$$OA = \frac{\sum_{i=0}^k p_{ii}}{\sum_{i=0}^k \sum_{j=0}^k p_{ij}}, \quad (11)$$

$$mAcc = \frac{1}{k+1} \sum_{i=0}^k \frac{p_{ii}}{\sum_{j=0}^k p_{ij}}, \quad (12)$$

where k denotes the category, i and j are denoted as the ground truth and the predicted value respectively. Parameters of the form p_{xy} denote the prediction of x to y .

In addition, ablation experiments are performed to verify the importance of analyzing each module of ADNet and robustness tests are done.

4.1. Datasets and implementation details

ScanNet v2 dataset. The dataset contains RGB-D data of more than 1513 indoor scenes and provides labeling information for 20 fine-grained object categories (e.g., sofa, TV, bookshelf, etc.) and 9 coarse-grained scene categories (e.g., office, kitchen, bedroom, etc.). The training, validation, and test sets consist of 1201, 312, and 100 scenes, respectively.

S3DIS dataset. The dataset contains six indoor areas such as offices and conference rooms, totaling 271 rooms, with area 5 serving as the test set for the experiment. It was collected through Matterport, which provides color information (RGB) and spatial coordinates (XYZ) and uses semantic labels to define 13 different categories to which each point belongs.

Table 2
Semantic segmentation results on the ScanNet v2 dataset.

Methods	Supervision ratios	mIoU/%
PSD (Zhang et al., 2021) (ICCV)	1%	54.7
HybridCR (Li, Xie et al., 2022) (CVPR)	1%	56.8
GaIA (Lee et al., 2023) (WACV)	1%	65.2
AADNet (Pan et al., 2023)	1%	66.8
Pan et al. (2024) (AAAI)	1%	67.4
SPH3D-GCN (Lei et al., 2021) (TPAMI)	Fully	61.0
SegGroup (Tao et al., 2022) (TIP)	Fully	62.7
APCF-Net (Lin et al., 2022) (GRSL)	Fully	63.1
LGGCM (Du et al., 2022) (TNNLS)	Fully	64.4
DCL (Yao et al., 2023) (RA-L)	Fully	64.4
LG-Net (Zhao, Ma et al., 2023) (CG)	Fully	66.5
BADENet (Zhao et al., 2024) (CAEE)	Fully	64.9
GDFNet (Lu et al., 2024) (OALIE)	Fully	60.6
ADNet (Ours)	Fully	67.6

Implementation details. For the ScanNet v2 dataset, we split the samples into blocks of size $2\text{ m} \times 2\text{ m} \times 3\text{ m}$ with 0.5 m as padding. Each block is then randomly sampled to obtain a point cloud of size 8192 as input to the network, where each point contains 3D spatial coordinates and RGB information. For the S3DIS dataset, we partitioned all the rooms into blocks with area $2\text{ m} \times 2\text{ m}$ and spacing of 0.5 m . Then 4096 points in each overlapped block are randomly selected as inputs to the network. We not only tested for area 5, but also used the same 6-fold cross-validation for all areas. All datasets are trained on RTX 4090 GPU with 200 epochs and batch_size 16. In addition, SGD is used to implement the model optimization, with the initial learning rate set to 0.001, momentum to 0.9, and dropout to 0.5.

4.2. Semantic segmentation on the ScanNet v2 dataset

Our proposed ADNet is evaluated on the test set of the ScanNet v2 dataset and compared with other recent work. These include weak supervision networks (PSD, HybridCR, GaIA, AADNet, and Pan et al.) and full supervision networks (SPH3D-GCN, SegGroup, APCF-Net, LGGCM, DCL, LG-Net, BADENet, and GDFNet). The quantitative results shown in Table 2 indicate that ADNet achieves better segmentation performance with the mIoU of 67.6%. Compared to SPH3D-GCN, the mIoU of this network is significantly improved by 6.6%. The network also achieves a slight improvement of 0.2% in mIoU compared to the best performing model (1%) proposed by Pan et al. In addition, ADNet's mIoU is improved by 2.7% compared to the latest BADENet, which is also fully supervised. In indoor scenes with objects of different sizes and shapes, the realized performance improvement is largely attributed to the fact that ADNet can capture both local details and global contextual information.

Moreover, we visualize some example segmentation comparison results of ADNet with DGCNN in Fig. 7. From the figure, we can clearly see that ADNet can segment objects of different scales more accurately even in complex indoor scenes.

For example, for the minor categories of furniture such as 'sofa' and 'cabinet' labeled by numbers 1 and 10, the ALNS in ADNet can dynamically adjust the neighborhood scale to capture their unique geometric features. Although there are still some errors in the visualization at the boundaries of the indoor building structure and furniture, such as walls and doors, clutter, beds, and other detailed parts, the overall segmentation quality has improved significantly. In addition, the segmentation accuracy of ADNet for the clutter can be further improved. For 'chair' and 'window' included in numbers 2, 3, 4, and 9, GSA-Conv and SPP-IPCs in ADNet can capture the spatial relationship between objects and utilize global contextual information to understand the interior layout for more accurate segmentation. For the three types of furniture ('table', 'desk', and 'bed') marked by numbers 3, 5, 6, 8, and 9, DCAA-Conv enables ADNet to focus on more important local features to distinguish these objects with planar structural similarities. It is worth mentioning

that ADNet can effectively avoid the noise effects caused by uneven lighting in indoor scenes, such as the furniture marked by labels 6, 8, 9, and 10, through the attention mechanism in the multiscale feature fusion module.

4.3. Semantic segmentation on the S3DIS dataset

In addition to evaluating our approach on the ScanNet v2 dataset, we also conducted experiments on the large-scale indoor scene S3DIS dataset. Specifically, we evaluate the semantic segmentation performance of ADNet on both 6-fold cross-validation and area 5 of the S3DIS dataset.

6-fold cross-validation of S3DIS. To evaluate the generalization performance of ADNet under different data distributions, we conducted a 6-fold cross-validation experiment. In Table 3, we list the quantitative results of ADNet and other networks, including PointNet++, SPG, DGCNN, KPConv, RandLA-Net, IAF-Net, SCF-Net, CAN, DenseKPNet, FA-ResNet, DCNet, VFEO, and HADF-Net. The experimental results show that the mIoU of ADNet reaches 72.9%, which is superior to all other networks, especially compared with PointNet++ and DGCNN, which increase by 18.4% and 16.8% respectively. Compared with CAN which also introduces context awareness, ADNet has a 4.6% improvement on mIoU. Although ADNet has slightly lower OA and mAcc than HADF-Net, its mIoU is 0.2% higher. This is because the proposed ALNS mechanism of ADNet can just solve the limitations of HADF-Net such as the loss of details brought by using traditional KNN for local neighborhood search. In addition, ADNet is 1.0% higher than DenseKPNet on mIoU, although its OA is slightly inferior by 0.4%. This further validates the effectiveness of ADNet in processing feature perception and semantic understanding of the whole indoor scene.

Some qualitative visualization results for all regions of the S3DIS dataset are shown in Fig. 8. It can be clearly seen that the proposed ADNet performs well in most categories of segmentation. In particular, compared to the classical graph convolutional network DGCNN, ADNet can recognize more accurately the segmentation of the 'column' shown in labels 1, 10, and 14. This is because the network performs the fusion of multi-scale features through SPP-IPCs, which enhances the ability to characterize the structure of the column. Moreover, ADNet has a significant visual improvement effect on the recognition accuracy of the minor categories of objects ('sofa' shown in labels 8 and 11).

Test on area 5 of S3DIS. Table 4 shows the semantic segmentation results of ADNet with other works on area 5, including the classical point-based processing networks (PointNet++ and KPConv), representative graph convolutional networks (HDGCN, SegGCN, SPH3D-GCN, LGGCM, and GDFNet), and recent new networks (IAF-Net, PSD, NeiEA-Net, FA-ResNet, and DRNet). We can see that ADNet achieves 68.7% and 88.8% mIoU and OA, respectively, outperforming all networks except JSNet++. The most obvious of them are 14.4% and 5.2% improvement over PointNet++, respectively. And compared to FA-ResNet which achieved the best segmentation accuracy in recent years, the mIoU of ADNet is slightly higher by 0.6%.

Compared with JSNet++, which also integrates features of different layers, the mIoU of ADNet is 10.7% higher, although the OA of ADNet is slightly inferior by 0.3%. This is mainly because ADNet can better adapt to multi-scale IPCs with different densities and irregular shapes. Compared to GDFNet with graph convolution, the mIoU of ADNet is 8.6% higher, although its OA is slightly inferior by 0.4%. This is because ADNet can fuse more comprehensive features while adaptively adjusting the local neighborhoods of regions with different point cloud densities.

It is worth mentioning that ADNet achieves the highest segmentation accuracy in six categories: ceiling, floor, column, window, board, and clutter. In addition, our method has advantages in segmentation accuracy mainly in the confusing indoor structures and minor categories. The mIoU is 37.9% on column, a category confusable with beam and wall. For the minority category of windows, the mIoU is 65.8%, while

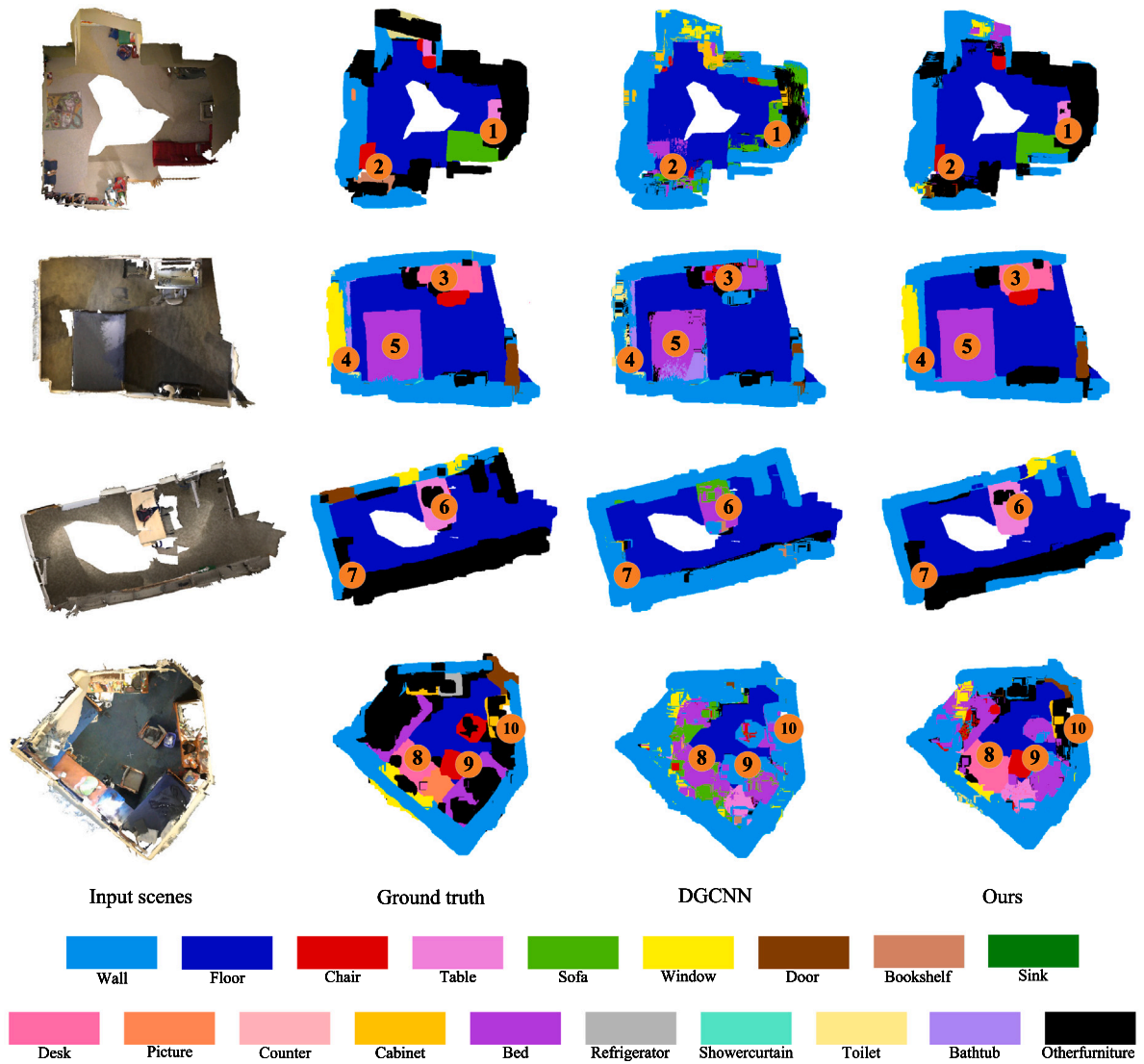


Fig. 7. Visual comparison on the ScanNet v2 dataset.

sofa, another minority category, achieves a mIoU of 71.3%, marking it as the second highest. This is due to two aspects: one is that the ALNS mechanism in ADNet effectively solves the problem of detail loss caused when extracting features with uneven density distribution, and the other is that the multi-scale feature fusion module enhances and learns the feature representation of different indoor areas.

Fig. 9 shows some qualitative results comparing with PointNet++ and DGCNN on the area 5 of S3DIS. In complex indoor environments, ADNet can more accurately distinguish between the door and the bookcase shown in label 1. This is because DCAA-Conv is able to focus on the semantic context information of both to understand the surrounding environment. By dynamically determining the local neighborhood size of each point, ALNS allows the network to more precisely capture the boundary features between different elements to distinguish other objects from walls ('board' and 'clutter' shown in labels 2, 5, and 6). In addition, the recognition of the columns shown in labels 3 and 7 shows some degree of visual improvement compared to both PointNet++ and DGCNN although there are still problems of mis-segmentation as well as incomplete segmentation compared to the ground truth. Compared with PointNet++, GSA-Conv in ADNet can better understand the shape and outline of the 'table' shown in label 8 by enhancing the feature representation capability of the point cloud, thus improving the segmentation accuracy of the object.

Table 3

Semantic segmentation results on the 6-fold cross-validation of S3DIS.

Methods	mIoU/%	OA/%	mAcc/%
PointNet++ (Qi, Yi et al., 2017) (NIPS)	54.5	81.0	67.1
SPG (Landrieu & Simonovsky, 2018) (CVPR)	62.1	85.5	73.0
DGCNN (Wang et al., 2019) (TOG)	56.1	84.1	—
KPConv (Thomas et al., 2019) (ICCV)	70.6	—	79.1
JSNet (Zhao & Tao, 2020) (AAAI)	61.7	88.7	71.7
RandLA-Net (Hu, Yang et al., 2020) (CVPR)	70.0	88.0	82.0
IAF-Net (Xu et al., 2021) (AAAI)	70.3	88.8	77.8
SCF-Net (Fan et al., 2021) (CVPR)	71.6	88.4	82.7
CAN (Liu, Zeng et al., 2022) (TGRS)	68.3	88.6	79.0
DenseKPNet (Li, Li et al., 2022) (TGRS)	71.9	89.3	79.7
FA-ResNet (Zhan et al., 2023) (JAG)	71.7	88.8	83.1
DCNet (Yin et al., 2023) (TCSVT)	72.4	89.3	82.1
VFE0 (Jiang et al., 2024) (SPL)	72.4	88.3	82.4
HADF-Net (Zhou et al., 2024) (PR)	72.7	89.2	83.7
ADNet (Ours)	72.9	88.9	82.4

4.4. Model complexity

To verify the superiority of ADNet, we evaluate the performance and efficiency of the proposed method using two important metrics: FLOPs



Fig. 8. Visual comparison on the S3DIS dataset (all areas).

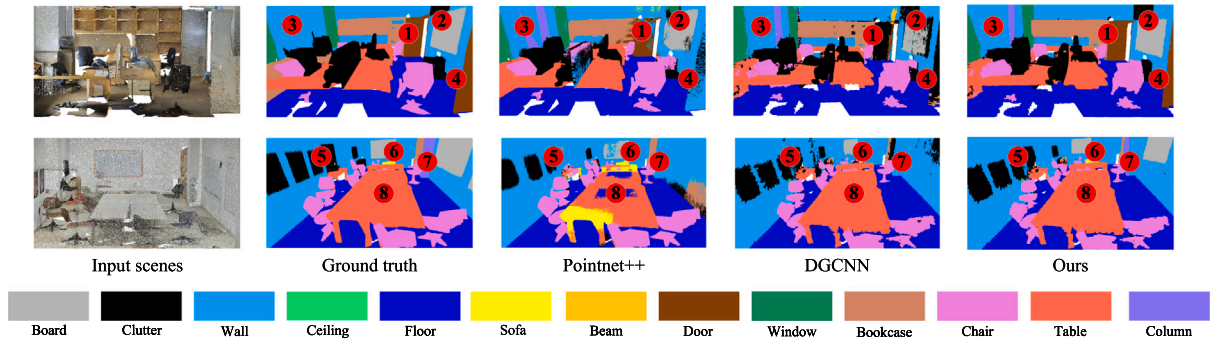


Fig. 9. Visual comparison on the S3DIS dataset (area 5).

(time complexity) and the number of parameters (space complexity). Taking area 5 of S3DIS as an example, we compare ADNet with some representative methods under the same experimental conditions, as

shown in Table 5. It is not difficult to find that the FLOPs of the proposed ADNet are higher than PointNet++ and LGCM. This is because ADNet needs to dynamically adjust the local neighborhood

Table 4
Semantic segmentation results on area 5 of S3DIS.

Methods	mIoU	OA	Ceil.	Floor	Wall	Beam	Colu.	Wind.	Door	Table	Chair	Sofa	Book.	Board	Clut.
PointNet++(NIPS) (Qi, Yi et al., 2017)	54.3	83.6	90.7	97.0	75.8	0.0	6.3	58.2	19.4	69.5	74.9	50.8	62.1	58.4	43.0
HDGCN(ICRA) (Liang et al., 2019)	59.3	–	94.0	98.3	80.3	0.0	24.0	48.2	33.8	77.7	86.9	46.3	71.1	57.0	53.8
KPConv(ICCV) (Thomas et al., 2019)	67.1	–	92.8	98.4	84.2	0.0	23.9	58.0	69.0	81.5	91.0	75.4	75.3	66.7	58.9
SegGCN(CVPR) Lei et al. (2020)	63.6	88.2	93.7	98.6	80.6	0.0	28.5	42.6	74.5	80.9	88.7	69.0	71.3	44.4	54.3
IAF-Net(AAAI) (Xu et al., 2021)	64.6	88.4	91.4	98.6	81.8	0.0	34.9	62.0	54.7	79.7	86.9	49.9	72.4	74.8	52.1
JEGCN(ICRA) (Han et al., 2021)	60.7	–	95.2	98.0	79.9	0.3	17.0	64.8	41.0	81.0	76.6	64.0	55.1	62.7	53.9
PSD(ICCV1) (Zhang et al., 2021)	65.1	–	92.3	97.1	80.7	0.0	32.4	55.5	68.1	78.9	86.8	71.1	70.6	59.0	53.0
LGGCM(TNNLS) (Du et al., 2022)	63.3	88.8	94.8	98.3	81.5	0.0	35.9	63.3	43.5	80.2	88.4	68.8	55.7	64.6	47.8
JSNet++(TCSVT) (Zhao & Tao, 2023)	58.0	89.1	93.7	98.5	80.5	0.0	16.9	57.2	41.9	76.8	84.7	30.5	60.2	58.3	54.9
NeiEA-Net(JAG) (Xu et al., 2023)	66.1	88.5	92.9	97.4	83.3	0.0	34.9	61.8	55.3	78.8	86.7	77.1	69.5	67.9	54.2
FA-ResNet(JAG) (Zhan et al., 2023)	68.1	88.9	94.0	97.9	82.6	0.0	39.1	64.1	65.2	78.5	87.5	77.6	69.8	73.3	56.0
DR-Net(TGRS) (Zhang & Bi, 2024)	64.2	88.7	93.1	97.3	81.8	0.0	22.5	56.0	65.8	79.8	87.1	57.1	73.2	63.5	57.6
GDFNet(OL) (Lu et al., 2024)	60.1	89.2	93.6	98.5	75.9	0.0	26.4	59.8	48.0	87.0	78.6	65.0	42.2	51.1	56.3
ADNet (Ours)	68.7	88.8	95.3	98.6	82.7	0.0	39.4	65.8	68.3	78.6	87.9	71.3	70.5	75.2	59.7

The units of mIoU and OA are in %.

Table 5
Spatio-temporal complexity of different methods on area 5 of S3DIS.

Methods	FLOPs/G	Parameters/M	mIoU/%	OA/%
PointNet (Qi, Su et al., 2017)	4.87	3.57	41.1	–
PointNet++(MSG) (Qi, Yi et al., 2017)	1.57	1.88	54.3	83.6
KPConv (Thomas et al., 2019)	–	14.93	67.1	–
SegGCN (Lei et al., 2020)	–	3.00	63.6	88.2
IAF-Net (Xu et al., 2021)	–	10.98	64.6	88.4
PSD (Zhang et al., 2021)	–	5.03	65.1	–
DGCNN (Wang et al., 2019)	8.42	0.99	56.5	87.0
LGGCM(with GSA) (Du et al., 2022)	1.87	4.26	63.3	88.8
NeiEA-Net (Xu et al., 2023)	–	4.89	66.1	88.5
FA-ResNet (Zhan et al., 2023)	–	18.20	68.1	88.9
BADNet (Zhao et al., 2024)	18.10	12.80	62.6	88.7
DR-Net(1%) (Zhang & Bi, 2024)	4.33	7.00	64.2	88.7
ADNet (Ours)	3.91	4.83	68.7	88.8

search range and consider more feature fusion when dealing with large-scale IPCs. ADNet with higher FLOPs can only be used in resource-rich devices, and running on some low-level mobile devices may reduce processing speed in real-time tasks. ADNet reduces redundant computations compared to DGCNN, which is also a graph convolutional neural network. This is due to the fact that ADNet avoids a fixed-size neighborhood search for each point, especially in sparse point clouds or regions with insignificant feature changes.

Meanwhile, ADNet has more parameters than the other five methods (PointNet, PointNet++, SegGCN, DGCNN, and LGGCM). This is because the DCAA-Conv and GSA-Conv in the model require additional trainable parameters to learn attention allocation, and the SPP-IPCs module also requires corresponding parameters to determine the pooling strategies at different levels. It also shows that ADNet still has some distance to become a highly efficient lightweight network, while there is a risk of overfitting when the amount of data is small, which can be avoided as much as possible by the dropout technique. Nonetheless, the mIoU of ADNet significantly outperforms other methods within acceptable time and computational cost. In addition, although the OA does not exceed FA-ResNet, the proposed ADNet has nearly four times fewer parameters than it. Therefore, we believe that ADNet strikes a balance between network complexity and segmentation accuracy.

To further verify the superiority of ADNet in inference time, we compare it with RandLA-Net, SCF-Net, and HADF-Net. To ensure the

fairness of the experiment, we set the batch_size of all models to 4 to better evaluate their generalization. Fig. 10 demonstrates that RandLA-Net possesses the fastest inference speed but has the lowest mIoU. In contrast, SCF-Net has the slowest inference. Our ADNet achieves highest mIoU and remains competitive with the latest HADF-Net. Comprehensive results show that ADNet achieves superior performance in acceptable inference times.

4.5. Ablation studies

To better verify the effectiveness of the ALNS, DCAA-Conv, GSA-Conv, and SPP-IPCs modules in the network, we evaluate their impact on segmentation accuracy by gradually replacing or removing each component. All experiments used 6-fold cross-validation to evaluate the S3DIS dataset. Tables 6 and 7 show the comparison results of the segmentation accuracy of the model by different neighborhood search mechanisms and different components of the multi-scale feature fusion module.

First, we keep the rest of the structure of ADNet unchanged and replace the proposed ALNS module with KNN, ball query, and octree in turn. As shown in Table 6, compared with our model, the mIoU and OA of the three models (a), (b), and (c) without the ALSN module decrease by 1.2%–4.0% and 0.4%–1.6%, respectively. This fully validates that

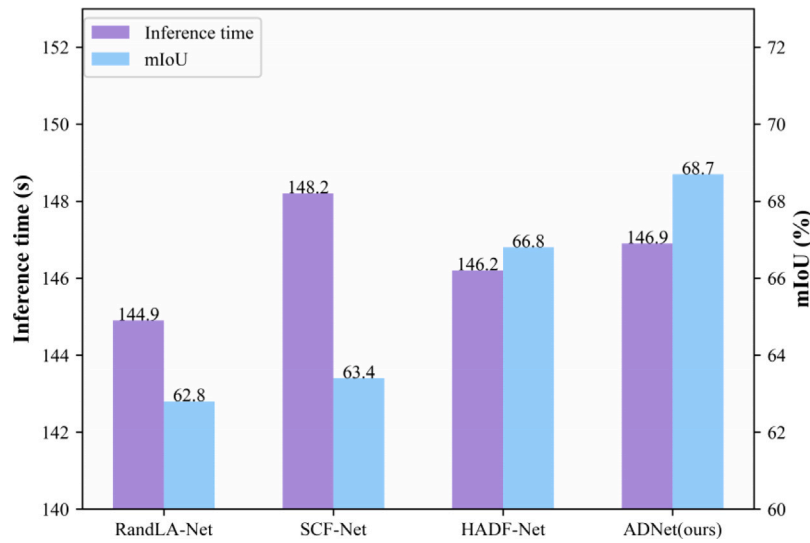


Fig. 10. Comparison of inference time on area 5 of S3DIS.

Table 6

The effectiveness of the ALSN module.

Models	Neighborhood search	mIoU/%	OA/%
(a)	KNN	68.9	87.3
(b)	Ball query	70.1	88.2
(c)	Octree	71.7	88.5
Ours	ALNS	72.9	88.9

the ALNS module can better adapt to the inhomogeneity of IPCs. Therefore, ADNet can effectively capture and extract local detail features in sparse or dense areas. Specifically, ALNS can mitigate the problem of insufficient neighborhood features caused by a fixed value of k in the sparse point cloud regions of model (a). In contrast to model (b), ALNS can ensure the number and quality of neighbors by dynamically adjusting the search scale. In addition, ALNS implements the search process based on the actual point cloud distribution, so it can effectively deal with the information loss caused by the hierarchical division of model (c).

We then adopt ALNS to construct the graph of the point cloud and leave the rest of the structure unchanged. One is to remove DCAA-Conv and GSA-Conv of the multi-scale feature fusion module successively, the other is to replace SPP-IPCs with the traditional max pooling (Max) and average pooling (Avg), respectively. As shown in Table 7, when the component structure of the module is incomplete, the segmentation accuracy of the model will be reduced to different degrees. For example, model (d) cannot effectively capture complex relationships between local features and global context due to the lack of DCAA-Conv, resulting in a decrease of mIoU and OA by 1.4% and 1.2%, respectively. The model (e) is unable to better understand spatial relationships between objects and the global layout of indoor scenes due to the absence of GSA-Conv, resulting in a 1.1% and 1.3% decrease in mIoU and OA for segmentation, respectively. Besides, our complete model improves mIoU by 2.2% and 1.9%, and OA by 0.6% and 0.8%, respectively, compared to the models (f) and (g) without SPP-IPCs. This is due to the ability of SPP-IPCs to efficiently aggregate features at different scales, allowing the model to capture both local details and global context information. In particular, SPP-IPCs preserves more context information through the combination of multiple pooling layers compared to traditional max pooling.

Overall, the advantages of ADNet in accurately identifying and segmenting point cloud data in indoor spaces of different sizes are as follows:

Table 7

The effectiveness of the multi-scale feature fusion module.

Models	DCAA-Conv	GSA-Conv	SPP-IPCs	mIoU/%	OA/%
(d)	×	✓	✓	71.5	87.7
(e)	✓	×	✓	71.8	87.6
(f)	✓	✓	(Max)	70.7	88.3
(g)	✓	✓	(Avg)	71.0	88.1
Ours	✓	✓	✓	72.9	88.9

- ALNS ensures that enough local information can be captured even in regions where the point cloud is sparse. This is particularly important when the network is applied to indoor robot navigation, especially when the robot recognizes environments with incomplete spatial information or obscured by obstacles.
- DCAA-Conv helps the network to distinguish regions with different contextual semantic information. In indoor space mapping, this module helps the system to more accurately identify and categorize different interior elements such as walls, furniture, etc.
- GSA-Conv applies an attention mechanism to enhance the ability of the network to perceive key points and edges in spatial structure. This will be very helpful for ADNet in indoor scene reconstruction to accurately understand the 3D structure.
- SPP-IPCs enables the network to aggregate both local details and global information at the same time, and to fuse features at different levels. The key information aggregated will ensure that the indoor scene reconstruction system generates more detailed 3D maps and models.

4.6. Robustness tests

To further validate the robustness of ADNet, we add random Gaussian noise as a disturbance to the input points. Then the DCAA-Conv and GSA-Conv in the multiscale feature fusion module are removed separately according to the idea of ablation experiment and compared with the complete network. LGGCM and GAC (Wang et al., 2019b) are also used as competing methods. For fairness, the input points of ScanNet v2 and S3DIS datasets are set to 8192 and 4096 in each network. As can be seen from Fig. 11, the attention mechanisms in both DCAA-Conv and GSA-Conv enable ADNet to focus on more critical features while ignoring less severe noise interference, and the robustness will be stronger, especially in the complete network. Meanwhile, the semantic

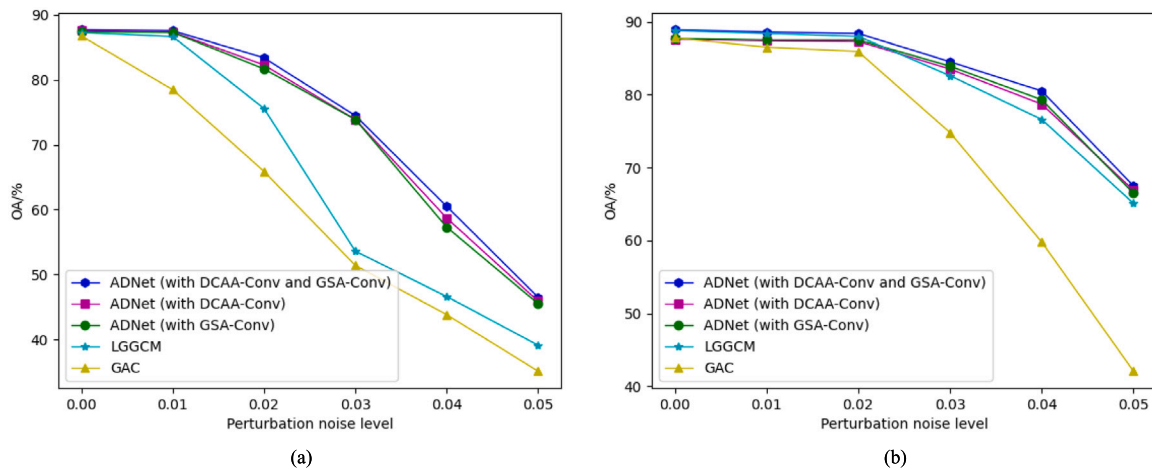


Fig. 11. Robustness test on indoor datasets. (a) ScanNet v2. (b) S3DIS.

segmentation accuracy of the proposed ADNet varies smoothly with increasing noise on both datasets. Some qualitative visualization results for all regions of the S3DIS dataset are shown in Fig. 10.

5. Conclusion

In this paper, a novel adaptive local neighborhood search and dual attention convolution network (ADNet) for complex semantic segmentation toward IPCs is proposed to improve the segmentation performance for recognizing objects and structures of different sizes in complex indoor environments. In contrast to most methods that use KNN for neighborhood search, we use ALNS in the network to dynamically adjust the local neighborhood radius for different point cloud densities and use it to construct the graph of IPCs to avoid the loss of local details. A multi-scale feature fusion module is also designed to effectively fuse local details, global contextual semantics, and spatial structure. In this module, SPP-IPCs utilize multi-window and overlapping pooling to ameliorate the problems of loss of details and degradation of spatial resolution that may result from common pooling operations. Comprehensive experimental results on two indoor datasets demonstrate that the proposed ADNet outperforms existing state-of-the-art (SOTA) methods: the mIoU value is 67.6% for ScanNet v2, and 68.7% and 72.9% for S3DIS in the area 5 and 6-fold cross-validation, respectively.

Limitations and future works. ADNet improves the semantic segmentation accuracy of IPCs through ALNS and multi-scale feature fusion modules. However, in more complex large-scale indoor scenes, we still need to spend a lot of point cloud labeling cost, and it will cause time consumption when the model does a more comprehensive local neighborhood search and feature fusion at more scales. In future work, we will exploit weak supervision to reduce the model's dependence on accurate labeling and time consumption. In addition, the efficiency of point cloud semantic segmentation can be further improved by introducing a more advanced self-attention mechanism (Transformer). In this way, we will achieve a better balance of accuracy and speed in our work and adaptively extend it to richer real-world indoor scenes, such as indoor robot navigation, indoor space mapping, and 3D scene reconstruction.

CRedit authorship contribution statement

Da Ai: Conceptualization, Formal analysis, Project administration, Methodology, Supervision, Funding acquisition, Writing – review & editing. **Siyu Qin:** Formal analysis, Investigation, Software, Methodology, Data curation, Writing – original draft. **Zihe Nie:** Software, Data curation, Validation, Writing – original draft. **Dianwei Wang:** Formal analysis, Validation. **Hui Yuan:** Writing – review & editing, Supervision. **Ying Liu:** Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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